FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Main)-2021

TEST DATE: 04-04-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. A

Sol. For pipe O, $\ell_0 = \lambda/2$ and for pipe C, $\ell_C = \lambda/4$, where λ is the wavelength corresponding to the fundamental mode. Therefore, $\lambda = 2\ell_0 = 4\ell_C$.

2. B

3. B

Sol. $\dfrac{\Delta Q}{Q} = \pm\left[n\dfrac{\Delta x}{x} + \dfrac{m\Delta y}{y}\right]$

4. A

Sol. $Q = \dfrac{\omega_0 L}{R} = \dfrac{\omega_0}{2\Delta\omega}$

$\therefore \quad 2\Delta\omega = R/L$

$2\pi(f_2 - f_1) = \dfrac{R}{L} \Rightarrow f_2 - f_1 = \dfrac{R}{2\pi L}$

5. B

Sol. After the switch is closed, the common potential

$V = \left(\dfrac{q_1'}{4\pi\varepsilon_0 r_1}\right) = \left(\dfrac{q_2'}{4\pi\varepsilon_0 r_2}\right)$ where $q_1' + q_2' = q_1 + q_2$ = total charge

6. A

Sol. $n \times 200 \times 10^6 \times 1.6 \times 10^{-19} = 10 \times 10^6$

$n = 3.125 \times 10^{17}\ sec^{-1}$.

7. C

Sol. there will be no acceleration in the system

8. C

9. D

Sol. $\lambda/2 = 1.5\ L$

$\Rightarrow \lambda = 3L = v/f$

$\Rightarrow f = v/3L$

10. D

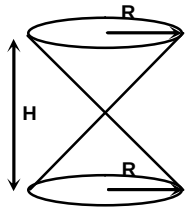
Sol. Both $\vec{V}$ and $\vec{B}$ are perpendicular to the plane of parallel wires

$\vec{F} = q(\vec{v} \times \vec{B}) =$ zero

11. C

12. C

Sol. The film will try to acquire minimum possible surface area which will be somewhat as shown in the figure. Thus net area will be less than $4\pi RH$. The further add credibility to the argument consider the surface shape shown in the figure. It can be easily shown that this surface area will be less than $4\pi RH$.



13. C

Sol. O_2(oxygen) N_2(nitrogen)

$nC_V(T_0 - T_1) = nC_V(T_2 - T_0)$

$T_0 = \frac{T_1 + T_2}{2}$

$T_0 = 300K$

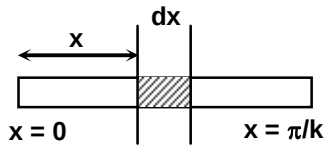
14. A

Sol. All others have cathode at higher potential than anode

15. C

Sol. Let us find the kinetic energy trapped between x = 0 and $x = \pi/k$ at $t = \pi/2\omega$

Consider the element shown



$v = \frac{\partial y}{\partial t} = -a\omega \sin(kx)\sin(\omega t)$

$dK = \frac{1}{2}(dm)v^2$

$\int dK = \int_0^{\pi/k} \frac{1}{2}\rho S a^2\omega^2 \sin^2 kx\ dx$

$= \frac{1}{4k}\rho S a^2\omega^2\pi$

This is the maximum kinetic energy (since at $t = \pi/2\omega$) the speed is maximum.

$\therefore$ this must also be the total mechanical energy.

16. C

Sol. ${}_w\mu_g < {}_a\mu_g$ and $\tan\theta = \mu$

17. A

Sol. Current $I = \dfrac{\sigma 2\pi r}{1/n} = \sigma 2\pi r n$

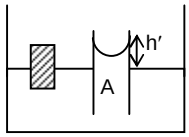
$$B = \frac{\mu_0 I r^2}{2(y^2 + r^2)^{3/2}} = \frac{\mu_0 I r^2}{2y^3} \text{ if } y \gg r$$

$$B = -\frac{\mu_0 r^2}{2y^3}(2\pi r\sigma n) = \frac{\mu_0 \pi r^3 \sigma n}{y^3}$$

18. A

Sol. There will be no change.

$\therefore$ $h' = h$



19. C

Sol. $x = A\sin(\omega t) \Rightarrow v = A\omega\cos(\omega t)$

$F = 6\pi\eta r v = 6\pi\eta RA\omega\cos(\omega t)$

$\Rightarrow$ Work done $= -\int F v dt = -6\pi\eta RA^2\omega^2 \int_0^{2\pi/\omega} \cos^2(\omega t)dt = -6\pi^2\eta RA^2\omega$

20. B

Sol. Refraction at convex lens.

$v_1 = 20$ cm

Refraction at concave lens $u = -10$ cm

$$\frac{1}{v} - \frac{1}{-10} = \frac{1}{-10} \Rightarrow v = -5 \text{ cm} \Rightarrow \text{height} = \frac{5 \text{ mm}}{2} = 2.5 \text{ mm} = 0.25 \text{ cm}$$

Hence the co-ordinates of the final image are (25 cm, 0.25 cm)

SECTION – B

21. 00000.60

Sol. $$4.5 = \frac{m_A C_A T_A + \frac{m_B C_B T_B}{2}}{m_A C_A + \frac{m_B C_B}{2}}$$

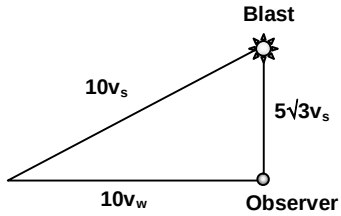
$$= \frac{m_B C_B}{m_A C_A} = 2$$

Final temperature

$$T = \frac{m_A C_A T_A + m_B C_B T_B}{m_A C_A + m_B C_B}$$

22. 00000.50

Sol. Location of epicentre of blast relative to ground after 10 seconds.



Let P be the location of the centre of sound wavefronts after 10 seconds. Then by geometry

$$(5\sqrt{3}v_s)^2 + (10v_w)^2 = (10v_s)^2$$

$$100v_w^2 + 75v_s^2 = 100v_s^2$$

$\Rightarrow$ $10\, v_w = 5v_s$

$$\frac{v_s}{v_w} = 2$$

$$\Rightarrow \quad \frac{v_w}{v_s} = \frac{1}{2} = 0.5$$

23. 00000.25

Sol. $$\frac{GMm}{R^2} = m\omega^2 R$$

$$\Rightarrow \quad GM = \omega^2 R^3$$

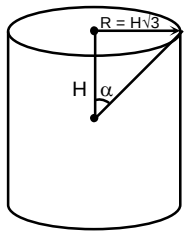
$$\Rightarrow \quad M \propto \frac{R^3}{T^2}$$

$$\Rightarrow \quad \frac{M_e}{M_p} = \left(\frac{R_e}{R_p}\right)^3 \cdot \left(\frac{T_p}{T_e}\right)^2 = 8 \times \frac{1}{4} = 2$$

$$\Rightarrow \quad \frac{x}{8} = \frac{2}{8} = 0.25$$

24. 00000.50

Sol. $$P = \frac{P_0}{4\pi} \times 2\pi\left(1 - \cos\frac{\pi}{3}\right)$$



$$P = \frac{P_0}{4} = \frac{\sigma A T^4}{4}$$

$$\eta = 4 \Rightarrow \frac{\eta}{8} = 0.50$$

25. 00000.75

Sol. $$hv = 13.6(3)^2\left[\frac{1}{4^2} - \frac{1}{5^2}\right] = 2.75 \text{ eV}$$

for n = 4 to n = 3

$$hv = (13.6) \times (3)^2 \left[\frac{1}{3^2} - \frac{1}{4^2}\right] = 5.95 \text{ eV}$$

for shorter wavelength

$3.95 = 5.95 - \phi \Rightarrow \phi = 2\text{eV}$

for longer wavelength

$eV_S = 2.75 - 2 = 0.75$ eV

26. 00000.66

Sol. $$\frac{1}{f_L} = \left(\frac{3}{2} - 1\right)\left(\frac{1}{20} + \frac{1}{20}\right)$$

$\Rightarrow f_L = 20$ cm

$\frac{1}{f}=\frac{1}{-10}-\frac{2}{20}=-\frac{1}{5}$

$\frac{1}{v}+\frac{1}{-30}=\frac{1}{-5}$

$\frac{1}{v}=\frac{1}{30}-\frac{1}{5}\Rightarrow v = -6$ cm

For lens

$\frac{1}{v'}-\frac{1}{-30}=\frac{1}{20}\Rightarrow v' = 60$ cm

The distance between the two images formed is x = 60 + 6 = 66 cm

27. 00002.50

Sol. $\frac{P_0^2}{2\rho V}=I=\frac{\text{Power}}{4\pi r^2}$

Put the values to get the answer

28. 00030.00

Sol. $10\sqrt{3}\cos 30^\circ=\frac{v}{2}$

v = 30 cm/s

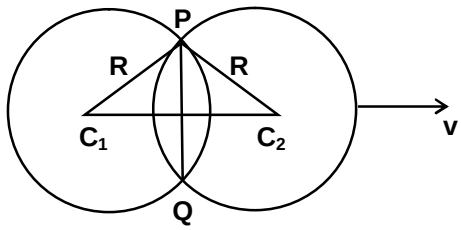
29. 00000.25

Sol. $\varepsilon = |(\vec{v}\times\vec{B})\cdot\vec{\ell}|$

ε = VB(PQ)

$\Rightarrow VB2\sqrt{R^2-\left(\frac{vt}{2}\right)^2}$

$=4\times0.25\times2\times3=6$ volts ($v_Q > v_P$)



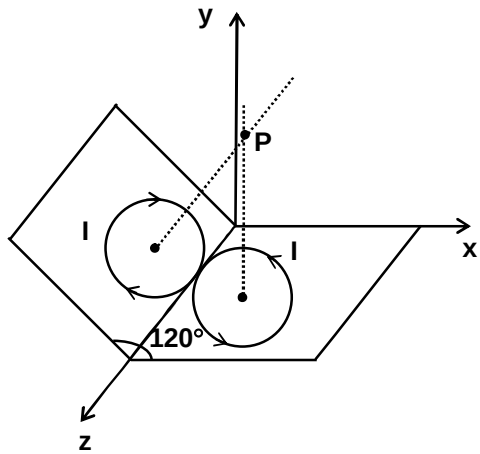
30. 00000.50

Sol. $B_1=\frac{\mu_0 I}{2}\frac{R^2}{(R^2+3R^2)^{3/2}}=\frac{\mu_0 I}{16R}$

$B^2=B_1^2+B_1^2+2B_1B_1\cos120^\circ$

$B=B_1=\frac{\mu_0 I}{16R}$

n = 0.50



Chemistry

PART – B

SECTION – A

31. C

Sol. $\frac{0.9}{180} \times 12 \times N_A = 0.06\ N_A$

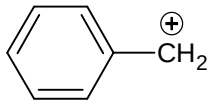
(A) $\frac{0.048}{32} \times 4 \times N_A = 0.006\ N_A$

(B) $\frac{0.17}{17} \times 3 \times N_A = 0.03\ N_A$

(C) $\frac{0.30}{30} \times 6 \times N_A = 0.06\ N_A$

(D) $\frac{0.03}{2} \times 2 \times N_A = 0.03\ N_A$

32. B

Sol. (A) $CH_2 = CH - \overset{+}{C}H_2$ and  are resonance stabilized carbocations, so these alcohols will give immediate turbidity.

(B) 1° alcohols gives fastest reactivity with Na metal but slowest reactivity in Lucas test.

(C) 1° alcohols gives slowest reactivity in Lucas test.

(D) It involves reaction between an acid (HCl) and a base (R – OH).

33. A

Sol. ZnS forms water soluble complex

$$ZnS + 4NaCN \longrightarrow 4Na^+ + \left[Zn(CN)_4\right]^{2-}$$

$\downarrow$

Water soluble

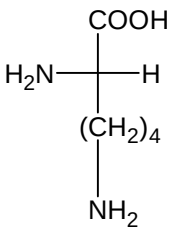
34. D

Sol. The electric arc is struck between the electrodes of metals immersed in the dispersion medium.

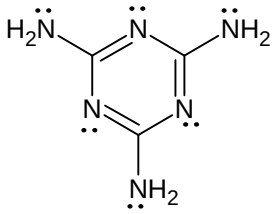
35. B

Sol. The other three are bacterostatic.

36. D

Sol. 

37. C

Sol. 

$C_3H_6N_6$

38. B

Sol. For bcc, $r = \frac{\sqrt{3}}{4}a$

Ede length not covered by atom = a – 2r

$= a - 2\left(\frac{\sqrt{3}}{4}\right)a = a\left(\frac{2-\sqrt{3}}{2}\right)$

% of edge length not covered $= \frac{a\left(\frac{2-\sqrt{3}}{2}\right)}{a} \times 100$

$= 0.134 \times 100 = 13.4\%$

39. C

Sol. Na_2CO_3 can remove permanent and temporary hardness of water by converting HCO_3^- and Cl^- / SO_4^{--} of Ca^{++} and Mg^{++} to precipitates.

40. C

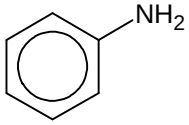
Sol. $Sb_2S_3 \Rightarrow Gr(II)$ Sulphide

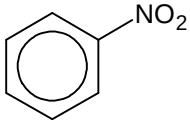
$MnS \Rightarrow Gr(IV)$ Sulphide

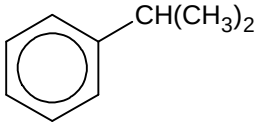
$BaS \Rightarrow Gr(V)$ Sulphide

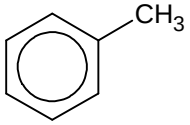
$K_2S \Rightarrow Gr(VI)$ Sulphide

41. B

Sol.  do not undergo F.C. reaction due to salt formation

 gives poor yields in F.C. since - NO_2 is a powerful deactivator

 $\Longrightarrow \alpha H = 1$ and +I slower rate

 $\Longrightarrow \alpha H = 3$ and +I fastest rate

42. C

Sol. CO_2 is evolved in all the cases, since it comes from $NaHCO_3$.

43. B

Sol. a, b vary with temp. and pressure and these constants have different values according to different equations for real gases and $T_B = \frac{a}{Rb}$ for real gases.

44. B

Sol. $\Rightarrow y_A = \frac{P_A^o x_A}{P_A^o x_A + P_B^o x_B}$

$\Rightarrow \frac{1}{y_A} = 1 + \left(\frac{P_B^o}{P_A^o}\right).\frac{x_B}{x_A}$

$\Rightarrow \frac{1}{y_A} = \left(\frac{P_B^o}{P_A^o}\right).\frac{1}{x_A} + \left(\frac{P_A^o - P_B^o}{P_A^o}\right)$

$\Rightarrow \frac{1}{x_A} = \left(\frac{P_A^o}{P_B^o}\right).\frac{1}{y_A} + \left(\frac{P_B^o - P_A^o}{P_B^o}\right)$

$\Rightarrow Y = MX + C$

$\text{Slope}(M) = \frac{P_A^o}{P_B^o}$

$y - \text{intercept} = \left(\frac{P_B^o - P_A^o}{P_B^o}\right)$

45. D

Sol. The complete hydrolysis equation is

$6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$

46. C

Sol. $\left.\begin{matrix} MnO_2 \\ Cr_2O_3 \end{matrix}\right\}$ Amphoteric

$\begin{cases} MnO \Rightarrow \text{Basic} \\ Mn_2O_3 \Rightarrow \text{Weakly basic} \\ Mn_2O_7 \Rightarrow \text{Strongly acidic} \end{cases}$

$\begin{cases} CrO \Rightarrow \text{Basic} \\ CrO_3 \Rightarrow \text{Acidic} \end{cases}$

47. C

Sol. H_2O is a weak field ligand, so cannot force pairing of electron(s)

$\left[Zn(H_2O)_6\right]^{++} \Rightarrow t_{2g}^6 e_g^4 \Rightarrow n = 0$

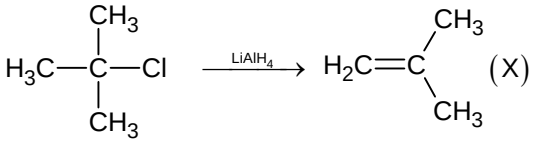
$\left[Cr(H_2O)_6\right]^{++} \Rightarrow t_{2g}^3 e_g^1 \Rightarrow n = 4$

$\left[Mn(H_2O)_6\right]^{++} \Rightarrow t_{2g}^3 e_g^2 \Rightarrow n = 5$

$\left[Cu(H_2O)_6\right]^{++} \Rightarrow t_{2g}^6 e_g^3 \Rightarrow n = 1$

48. B

Sol. LAH causes elimination in 3° alkyl halides.

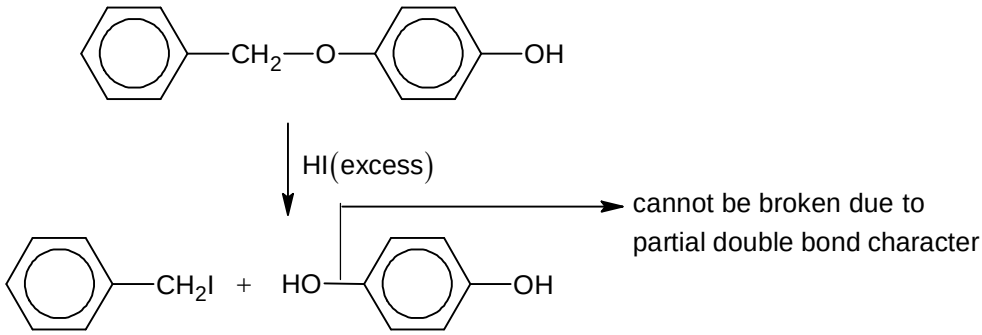


Ph_3SnH, i.e. TPH selectively reduces alkyl halides ($1^o/2^o/3^o$) but not C = C or C ≡ C

$CH_3CH=CHCH_2Cl \xrightarrow{Ph_3SnH} CH_3CH=CH-CH_3\,(Y)$

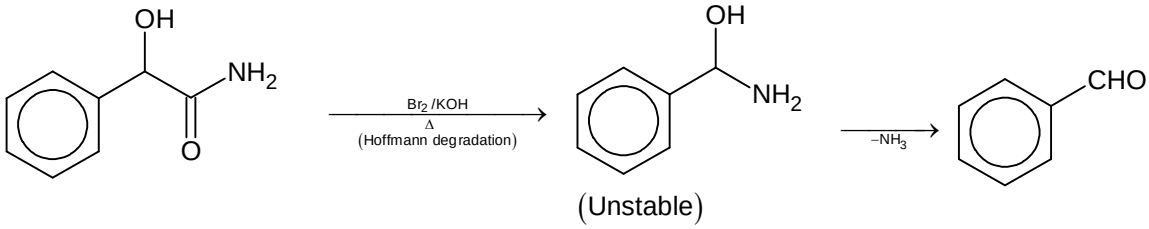
49. B

Sol.



50. B

Sol.



SECTION – B

51. 00004.00

Sol.

$X(g) \longrightarrow Y(g) + \frac{1}{2}Z(g)$

$150-P \quad P \quad P/2$

150 – P + P + P/2 = 170

P = 40 mm

So pressure of X (g) decreases by 40 mm in 10 min.

Rate of disappearance $= \frac{40}{10} = 4$ mm / min

52. 00032.65

Sol. $Cl^- \longrightarrow ClO_4^-$

$-1 \quad +7$

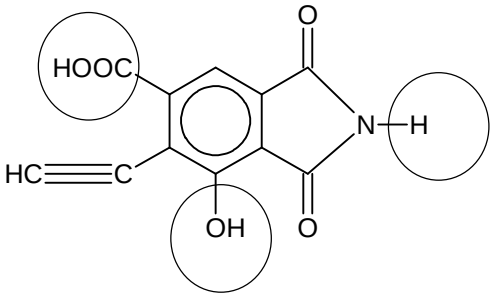
Number of Faraday = Number of equi. of $NaClO_4$ formed

$= \frac{500}{122.5} \times 8 = \frac{4000}{122.5}$

= 32.65

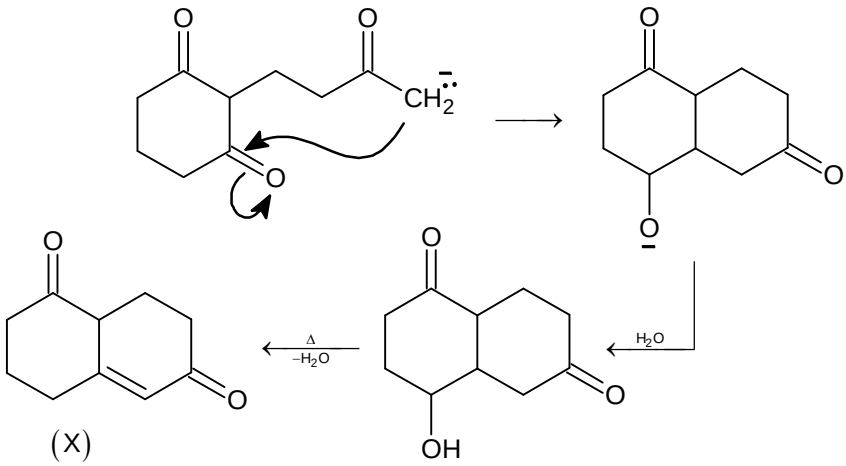
53. 00003.00

Sol.



54. 00005.00

Sol.

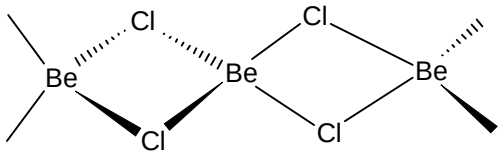


$\Rightarrow DU = \frac{20+2-12}{2} = 5$

i.e. 2 Rings + 3 double bonds.

55. 00004.00

Sol.



(Polymeric in solid state)

56. 00010.00

Sol. For triatomic non linear ideal gas

$C_V = 3\times\frac{R}{2}+3\times\frac{R}{2}+3\times R = 6R$

$C_m = C_V + \frac{R}{1-x}$

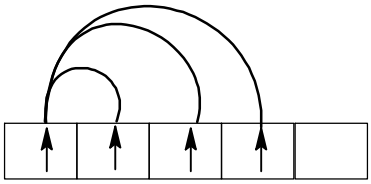
$C_m = 6R + \frac{R}{1-2} = 5R$

$\therefore C_m = 5\times 2$

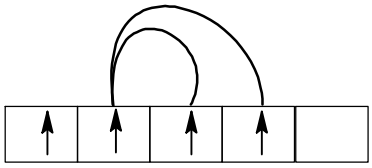
$C_m = 10 \text{ cal mol}^{-1}$

57. 00001.50

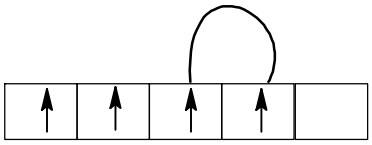
Sol.



3-Exchanges



2-Exchanges



1-Exchanges

Total 6 exchanges

$\frac{x}{4}=\frac{6}{4}=1.50$

58. 00012.50

Sol.

$$XY(g)\rightleftharpoons X(g)+Y(g)$$

Equi.	1	0	0	
Equi.	$1-\alpha$	α	α	$=1+\alpha$
Equi. pressure	$\left(\frac{1-\alpha}{1+\alpha}\right)P$	$\left(\frac{1-\alpha}{1+\alpha}\right)P$	$\left(\frac{1-\alpha}{1+\alpha}\right)P$	

$$K_P=\frac{\left(\frac{\alpha}{1+\alpha}\right)P.\left(\frac{\alpha}{1+\alpha}\right)P}{\left(\frac{1-\alpha}{1+\alpha}\right)P}$$

$$K_P=\frac{\alpha^2}{1-\alpha^2}.P$$

$$\Rightarrow \frac{K_P}{P}=\frac{(1/3)^2}{1-\left(\frac{1}{3}\right)^2}=\frac{1}{9}\times\frac{9}{8}=\frac{1}{8}$$

$$\Rightarrow \frac{K_P}{P}=\frac{100}{8}\times10^{-2}=12.50\times10^{-2}$$

59. 00005.00

Sol. KNO_3 do not affect the solubility of AgCl

$$AgCl\rightleftharpoons Ag^{+}+Cl^{-}$$

$$S \quad S$$

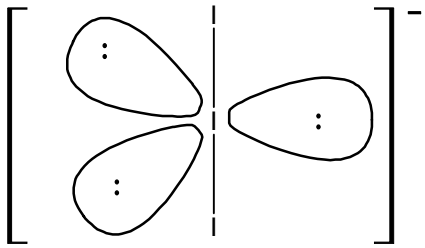
$$K_{SP}=S^2$$

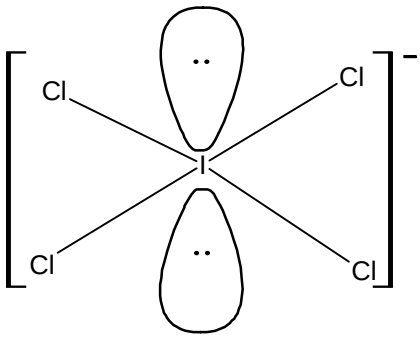
$$S=\sqrt{10^{-10}}=10^{-5}\ mol/L^{-1}$$

$$\Rightarrow -\log S=5.00$$

60. 00003.50

Sol.





$x = 3 \times 2 = 6 \qquad y = 4 \times 2 = 8$

$\therefore \dfrac{x+y}{4} = \dfrac{8+6}{4} = 3.50$

Mathematics

PART – C

SECTION – A

61. D
Sol. $x^2 = t$; $|t - 9| + |t - 5| \geq 4$

62. D
Sol. $[x] + \left[x + \frac{1}{2}\right] = [2x]$

63. C
Sol. Distance between (0, 2) and (0, –4) = 6 < 10
PS + PS′ = 2a > SS′

64. A
Sol. Least degree term = 0
$\Rightarrow 2x - y = 0$

65. A
Sol. $(ab)^7 (ac)^3$ is $(ab + bc + ca - 2)^{10} = {}^{10}C_7$

66. D
Sol. $\sqrt{3}\,\text{cosec}\,20° - \sec 20° = 4$

67. B
Sol. Let $\ln\frac{\sqrt{4-x^2}}{1-x} = \theta$

$\because\ 0 < \frac{\sqrt{4-x^2}}{1-x} < \infty \Rightarrow \ln\left(\frac{\sqrt{4-x^2}}{1-x}\right) \in (-\infty, \infty)$

68. D
Sol. $[x]\sin^2(\pi x)$ is differentiable $\forall\ x \in R$

69. A
Sol. $P(E) = P(P_1)P\left(\frac{E}{P_1}\right) + P(P_2)P\left(\frac{E}{P_2}\right) = \frac{1}{2}\times\frac{4}{7} + \frac{1}{2}\times\frac{6}{10}$

70. A
Sol. $\frac{x^2 + x - 5}{(x-1)^2} = 1 + \frac{3}{x-1} - \frac{3}{(x-1)^2}$

71. D
Sol. $|2z_1 - 3z_2 - 4z_3| = |2z_1 - 3(z_2 - 2) - 4(z_5 - 5) - 26| \leq 2|z_1| + 3|z_2 - 2| + 4|z_3 - 5| + 26$

72. B
Sol. Coefficient of x^{20} in $(1-x)^{-1}(1-x)^{-1}(1-x)^{-1}(1-x)^{-1}$

Coefficient of x^{20} in $(1-x)^{-3}(1-x^4)^{-1} = 536$

73. A

Sol. $\frac{n^2-2n}{\lfloor n}-\frac{n-1}{\lfloor n}=\frac{n-2}{\lfloor n-1}-\frac{(n-1)}{n(n-1)\lfloor n-2}=\frac{n-2}{(n-1)(n-2)\lfloor n-3}-\frac{1}{n\lfloor n-2}=\frac{1}{(n-1)\lfloor n-3}-\frac{1}{n\lfloor n-2}$

74. D

Sol. $(a-2b)^2+(b-2c)^2=0$

75. A

Sol. $\frac{\frac{a}{b}-1}{\frac{a}{b}+1}=\tan\left(\frac{A-B}{2}\right)\tan\left(\frac{C}{2}\right)\Rightarrow\tan\left(\frac{A-B}{2}\right)=1$

$A-B=90°$

76. A

Sol. $x=0$ and $x+y=0$

77. A

Sol. General points on lines are $P(\lambda+1, 2\lambda+2, 3\lambda+3)$ and $Q(3\mu+2, \mu+3, 2\mu+1)$

Let plane be $ax+by+cz=1$

Mid-point of PQ will lie on the plane for all values of λ and μ

$\Rightarrow a\left(\frac{3\mu+\lambda+3}{2}\right)+b\left(\frac{2\lambda+\mu+5}{2}\right)+c\left(\frac{3\lambda+2\mu+4}{2}\right)=1$

$\Rightarrow 3a+b+2c=0,\ a+2b+3c=0,\ 3a+5b+4c=2$

$\Rightarrow a=\frac{1}{9}, b=\frac{7}{9}, c=-\frac{5}{9}$

78. B

Sol. Draw the graph of $y=\left|\sin^{-1}(\sin x)\right|$ and $y=\left(\frac{\pi-x}{2}\right)$

79. D

Sol. $\sim(p\to(q\wedge r))\equiv p\wedge\sim(q\wedge r)\equiv p\wedge(\sim q\vee\sim r)$

80. B

Sol. Let $(x_i-8)=d_i$

So, $6x=6d=\sqrt{\frac{\sum d_i^2}{n}-\left(\frac{\sum d_i}{n}\right)^2}=\sqrt{\frac{45}{18}-\left(\frac{1}{2}\right)^2}=\frac{3}{2}$

SECTION – B

81. 00001.00

Sol. $x\left\{\left(1+\frac{1}{x}\right)^{\frac{1}{6}}-\left(1-\frac{1}{x}\right)^{\frac{1}{6}}\right\}=x\left\{1+\frac{1}{6}\frac{1}{x}+0\left(\frac{1}{x}\right)-\left(1-\frac{1}{6}\frac{1}{x}+0\left(\frac{1}{x}\right)\right)\right\}=\frac{1}{3}$

82. 02021.00

Sol. $A^{2021}=\begin{pmatrix}1 & 0\\ 4042 & 1\end{pmatrix}$

83. 00000.00

Sol. $A=\begin{pmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{pmatrix}$, $B=\begin{pmatrix} b_1 & b_2 \\ b_3 & b_4 \\ b_5 & b_6 \end{pmatrix}$

$$|BA|=\begin{vmatrix} a_1b_1+b_2a_4 & __ & __ \\ __ & __ & __ \\ __ & __ & __ \end{vmatrix}=\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ 0 & 0 & 0 \end{vmatrix}\times\begin{vmatrix} & & \\ & & \\ & & \end{vmatrix}=0$$

84. 00003.00

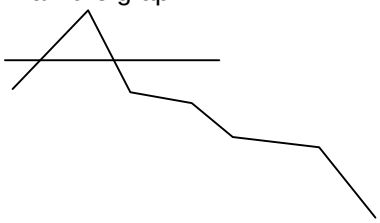
Sol. $d\left(\frac{y^2}{x}\right)=d\left(x-\frac{1}{x}\right) \Rightarrow \frac{y^2}{x}=x-\frac{1}{x}+c \Rightarrow y^2=x^2-1+cx$

$x = 1, y = 0; c = 0$

$\therefore y^2 = x^2 - 1$

85. 00001.00

Sol. Draw the graph



86. 00001.00

Sol. $y' = x^2 - 3x + 2$

Function $\uparrow$ on $(-\infty, 1)$ and $\downarrow$ on $(1, 2)$ then $\uparrow$ on $(2, \infty)$

87. 00000.50

Sol. $(\bar{A}\cdot\bar{B})\bar{R}-(\bar{A}\cdot\bar{R})\bar{B}=(\bar{A}\cdot\bar{B})\bar{C}-(\bar{A}\cdot\bar{C})\bar{B}$

$\Rightarrow -38\bar{R}-\bar{O}=-38\bar{C}-19\bar{B}$

88. 00002.00

Sol. $(\bar{a}+\bar{b})\cdot((\bar{b}+\bar{c})\times(\bar{c}+\bar{a}))=(\bar{a}+\bar{b})\cdot(\bar{b}\times\bar{c}+\bar{b}\times\bar{a}+\bar{c}\times\bar{a})$

89. 00008.00

Sol. $\frac{(4.2)(3.1)}{9.7.5.3.1}=\frac{8}{315}$

90. 00008.59

Sol. $\begin{vmatrix} -1 & 5 & -3 \\ -4 & 3 & 3 \\ 1 & 7 & k+1 \end{vmatrix}=0$, $k=-\frac{146}{17}$